

Lab Program 7

Title: Deploy a Web Application to Kubernetes using Minikube

Prerequisites:

Make sure Minikube, Kubectl are installed

STEP 1: Start Minikube

\$ minikube start

```
victus% minikube start
🐸 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚧 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
```

STEP 2: Check the status of Minikube

\$ minikube status

STEP 3: Create a deployment YAML file

\$ nano **nginx-deployment.yaml**

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

STEP 4: Create a deployment using nginx-deployment.yaml

```
$ kubectl apply -f nginx-deployment.yaml
```

```
victus% kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

STEP 5: Check the status of the deployment and the pod

```
$ kubectl get deployments
```

```
$ kubectl get pods
```

```
victus% kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    0/2     2            0           16s
```

```
victus% kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-pprxm  1/1     Running   0           98s
nginx-deployment-6f9664446b-rj4pl  1/1     Running   0           98s
```

STEP 6: To access the container externally, expose the port using the NodePort service

```
$ kubectl expose deployment nginx-deployment --type= NodePort --port=80
```

```
victus% kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
```

STEP 7: Check the status of service

```
$ kubectl get svc
```

```
victus% kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>       443/TCP          10h
nginx-deployment    NodePort    10.102.9.255  <none>       80:30298/TCP     8s
```

STEP 8: Access the web App

```
$ minikube service nginx-deployment
```

```
victus% minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30298

```
🌐 Opening service default/nginx-deployment in default browser...
```

The application will open in the default browser

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

STEP 9: Delete service and deployment

\$ kubectl delete svc nginx-deployment

```
victus% kubectl delete svc nginx-deployment
service "nginx-deployment" deleted
```

\$ kubectl delete deployment nginx-deployment

```
victus% kubectl delete deployments nginx-deployment
deployment.apps "nginx-deployment" deleted
victus% kubectl get pod
No resources found in default namespace.
victus% kubectl get deployments
No resources found in default namespace.
```